

Calculus A

Chapter 1: Functions

Section 1.1: Functions and Their Graphs

Main Lecture Notes

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Reference: Thomas' Calculus, 12th Edition

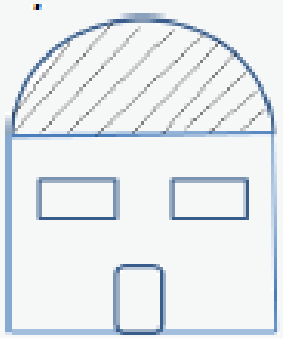
What is Calculus?

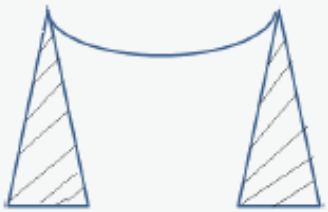
What is Calculus?

1. Calculus is the study of advanced algebra and geometry.
2. Calculus is the mathematics of change.

Example1

1. Calculate the rate of growth of a bacterial culture when different variables such as temperature and food sources are changed.
2. Determine the amount of material necessary to construct a curved dome over a new sports arena.
3. Determine the length of the cable hung between two towers that has shape of a catenary.





What is Calculus?

During the course we will see

1. How to turn a real life problem into a mathematical problem.
2. Functions which are used to turn problems into mathematical problems.
3. Instantaneous change of various functions (Differentiation) **Ex: velocity of a car**
4. Area and volume of certain regions (Integration)

Note

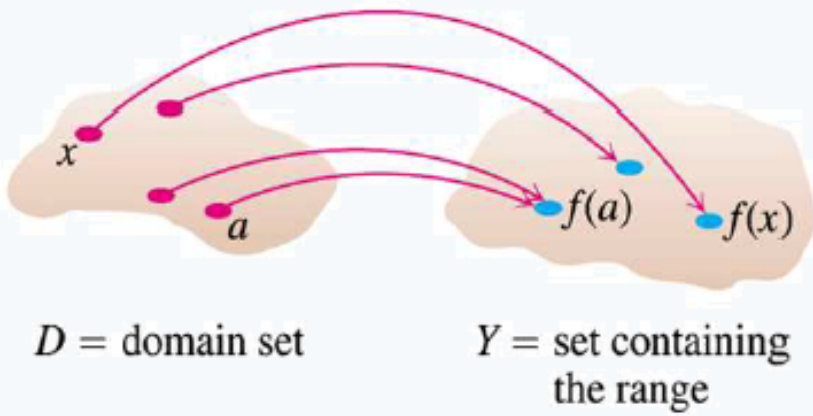
We start by assuming that you are familiar with the idea of a "set" and the set theoretic symbol "$\in$" ("an element of").

Definition

A function f from a set D to a set Y is a rule that assigns a unique (single) element $f(x) \in Y$ to each element $x \in D$.

D is called the **domain** of the function, Y is called the **co-domain** of the function. $f(x)$ is the value of f at x and is read as 'f of x'. x is called **independent** variable and $f(x) = y$ is called the **dependent** variable.

The set of all possible values of $f(x)$ as x varies throughout the domain is called the **range** of f. We can denote it by $\{f(x) | x \in D\} \subset Y$.



Finding Domain and Range of functions

Example

Find the domain and range of the following functions

1. Let $x \in \mathbb{R}, f(x) = x^2$ has domain $= \mathbb{R}$, and range $= [0, \infty)$.
2. Let $x \in \mathbb{R}, f(x) = \sqrt{1 - x^2}$ has domain $\mathcal{D}_f = [-1, 1]$, and range, $\mathcal{R}_f = [0, 1]$. **Explain how !**
3. Let $x \in \mathbb{R}, f(x) = \frac{1}{x-2}$ has domain $\mathcal{D}_f = (-\infty, 2) \cup (2, \infty)$, and range, $\mathcal{R}_f = \mathbb{R} \backslash \{0\} = (-\infty, 0) \cup (0, \infty)$. **Explain how !**

Domain Restrictions

Note

When finding domains, we usually look for bad values of x which must be excluded from the domain. Primary among the restrictions of the mathematical world is the fact that **you cannot now, nor in the future, divide by 0!!!** I will not divide by 0, my colleagues will not divide by 0, and you will not divide by 0.

In addition, in this class, we restrict our attention to real numbers. Therefore **we will not take square roots of negative numbers!**

This is a restriction which is rather different from the restriction against dividing by 0. There are areas of math "out there" in which people take square roots of negatives (in the world of complex numbers—in fact, in this world you can also take a logarithm of negatives and an inverse sine of numbers greater than 1). The course Complex Variables deals with this topic and it could be described as "the calculus of complex numbers."

Domain and Range Practice

Exercise

Find the domain and range of the following functions:

#	Function	Domain	Range
1	$f(x) = x^2 - 1$	$\mathcal{D}_f =$ _____	$\mathcal{R}_f =$ _____
2	$f(x) = \sqrt{x^2 - 9}$	$\mathcal{D}_f =$ _____	$\mathcal{R}_f =$ _____
3	$f(x) = \sqrt{x^2 - 3x}$	$\mathcal{D}_f =$ _____	$\mathcal{R}_f =$ _____
4	$f(x) = \dfrac{\lvert x \rvert}{x}$	$\mathcal{D}_f =$ _____	$\mathcal{R}_f =$ _____
5	$f(x) = x + \lvert x \rvert$	$\mathcal{D}_f =$ _____	$\mathcal{R}_f =$ _____
6	$f(x) = \sqrt{x} - 3$	$\mathcal{D}_f =$ _____	$\mathcal{R}_f =$ _____

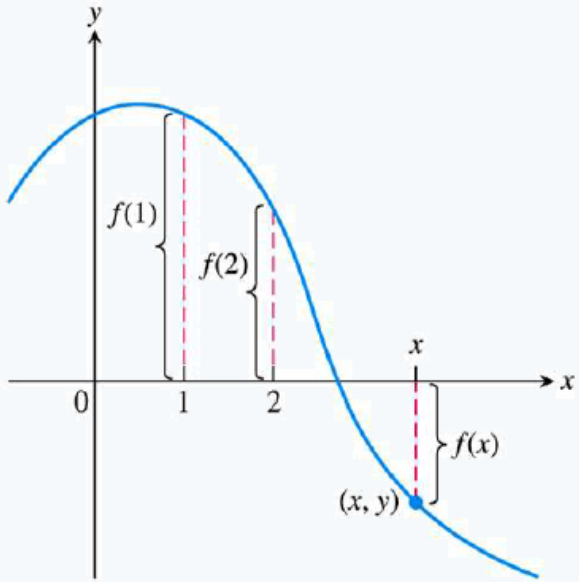
Key Solutions

#	Function	Domain	Range
1	$f(x) = x^2 - 1$	$\mathcal{D}_f = \mathbb{R}$	$\mathcal{R}_f = [-1, \infty)$
2	$f(x) = \sqrt{x^2 - 9}$	$\mathcal{D}_f = (-\infty, -3] \cup [3, \infty)$	$\mathcal{R}_f = [0, \infty)$
3	$f(x) = \sqrt{x^2 - 3x}$	$\mathcal{D}_f = (-\infty, 0] \cup [3, \infty)$	$\mathcal{R}_f = [0, \infty)$
4	$f(x) = \dfrac{\lvert x \rvert}{x}$	$\mathcal{D}_f = \mathbb{R} \setminus \{0\}$	$\mathcal{R}_f = \{-1, 1\}$
5	$f(x) = x + \lvert x \rvert$	$\mathcal{D}_f = \mathbb{R}$	$\mathcal{R}_f = [0, \infty)$
6	$f(x) = \sqrt{x} - 3$	$\mathcal{D}_f = [0, \infty)$	$\mathcal{R}_f = [-3, \infty)$

Representations of Functions

Definition

If f is a function with domain D, its graph consists of the points in the Cartesian plane whose coordinates are the input-output pairs of f. In set notation, the graph is $\{(x, f(x)) | x \in D\}$.



Vertical Line Test

Note

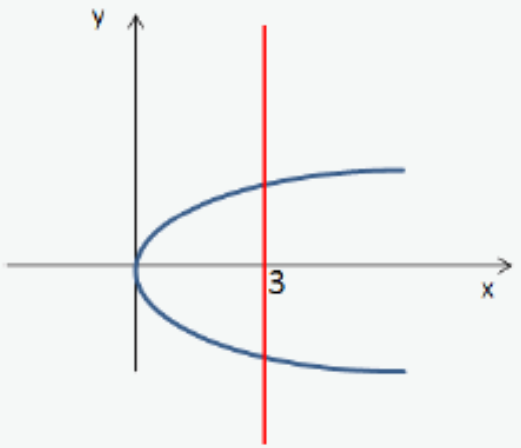
We do not have the converse statement, that is any curve may not be a function.

Vertical Line Test

A curve in the $xy-$plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.

Example

The blue curve below cannot represent a function as $x = 3$ has two different values assigned by the curve.



Piecewise Defined Function

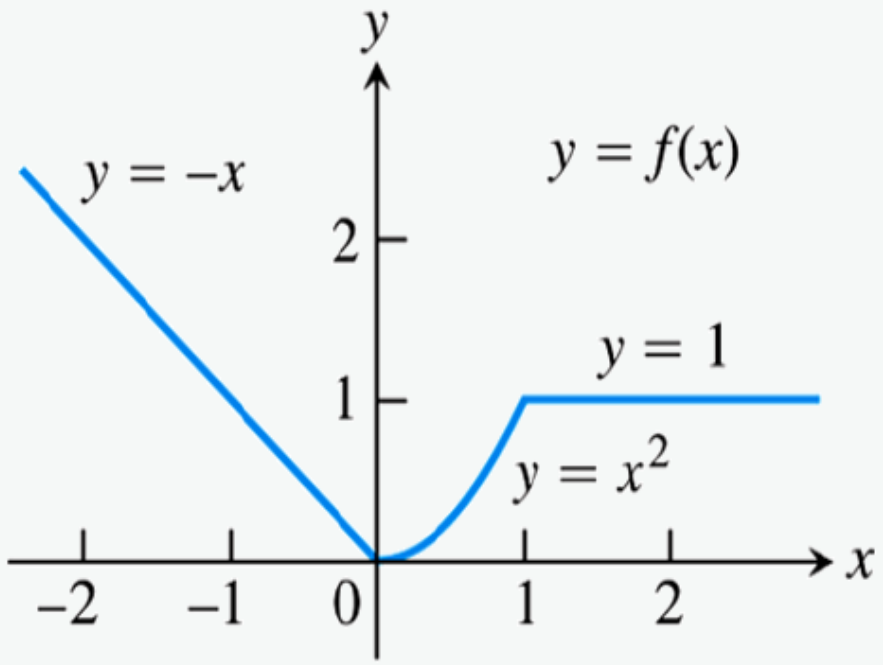
Note

Note. In this class, we will often define a function piecewise. That is, instead of giving a single formula for a function, we will give several formula which define the function piecewise over certain points or intervals. Though there may be several pieces, we will have only one function.

Example

$$f(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

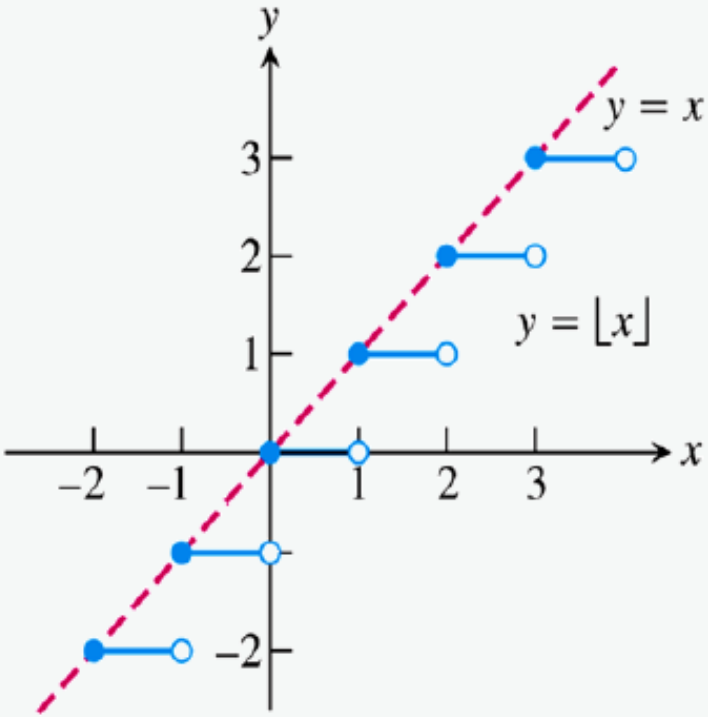
The graph of this function is:

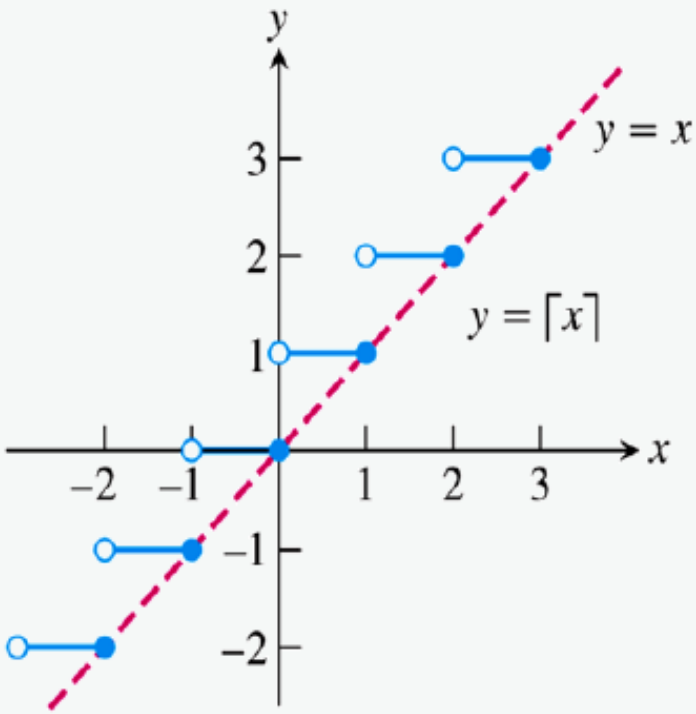


Piecewise Defined Function

Example

Two functions which will make useful examples in our study of one-sided limits (Section 2.4) are the greatest integer function $f(x) = \lfloor x \rfloor$ and the least integer function $g(x) = \lceil x \rceil$:





Increasing and Decreasing Functions

Definition

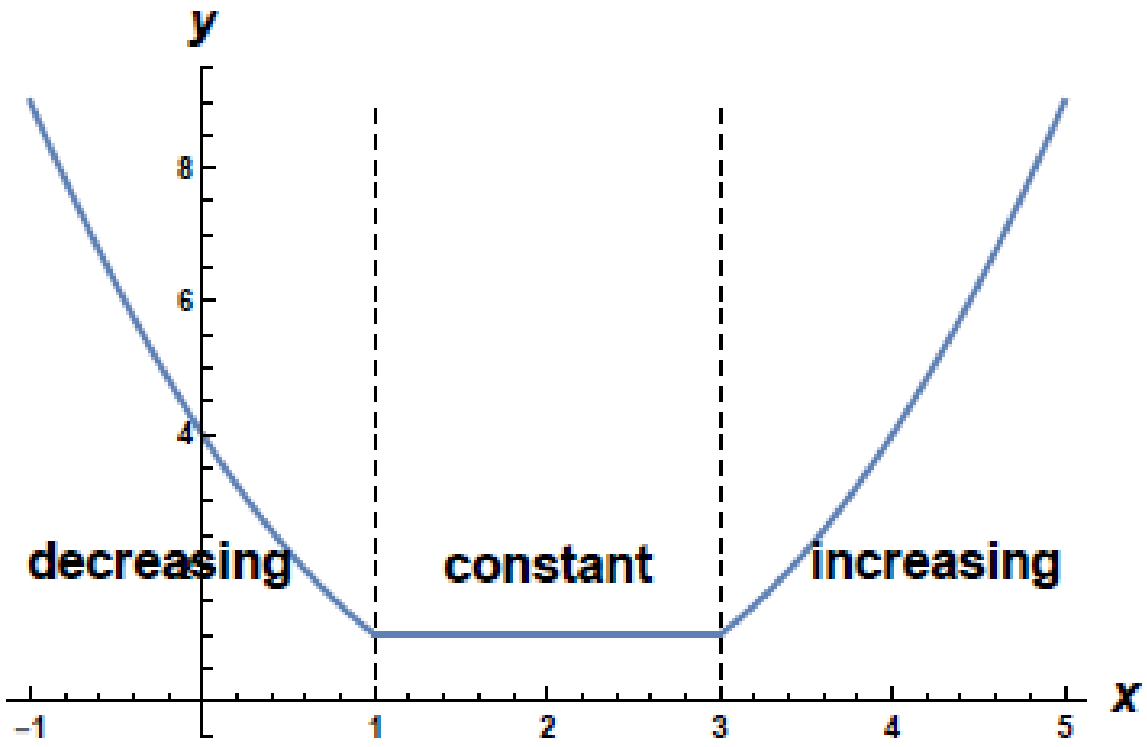
Let f be a function defined on an interval I and let x_1 and x_2 be any two points in I.

1. If $f(x_2) > f(x_1)$ whenever $x_1 < x_2$, then f is said to be increasing on I.
2. If $f(x_2) < f(x_1)$ whenever $x_1 < x_2$, then f is said to be decreasing on I.

Note

It is difficult to tell whether a function is increasing or decreasing unless you have the graph of the function. In Section 4.3 we will have a method to determine the increasing/decreasing properties of a function and then use these properties to create a graph.

Increasing and Decreasing Functions



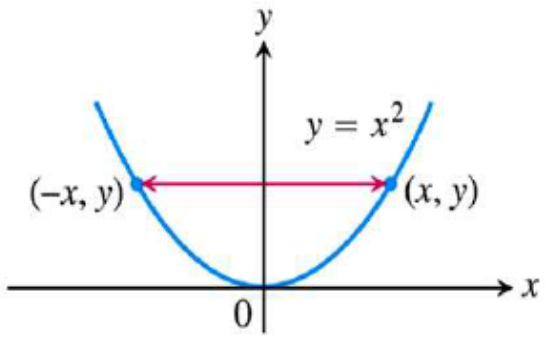
Even and Odd Functions

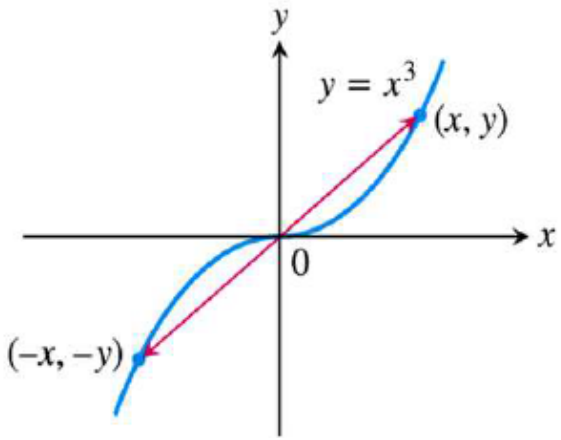
Definition

A function $y = f(x)$ is an

1. **even** function of x if $f(-x) = f(x)$,
2. **odd** function of x if $f(-x) = -f(x)$,

for every x in the function's domain.





The graph of an **even** function is said to be **symmetric** about the $y-$axis.
The graph of an **odd** function is said to be **symmetric** about the origin.

Even and Odd Functions

Example

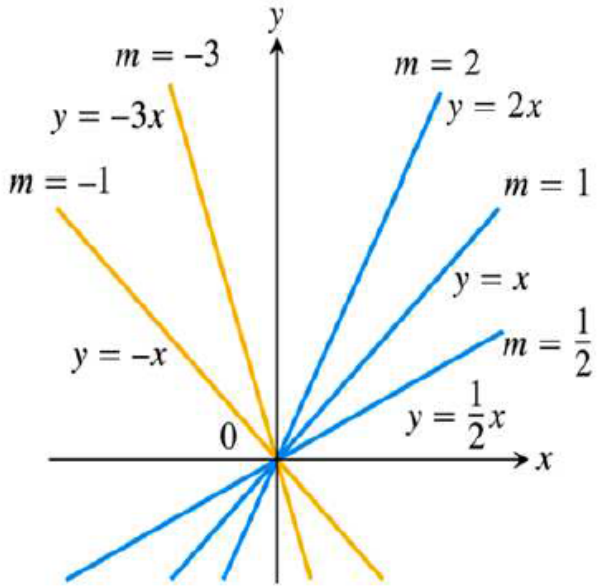
1. $f(x) = x|x|$ is an odd function.
2. $f(x) = \frac{x^2}{x^4+1}$ is an even function.
3. $h(t) = \frac{1}{t-1}$ not an even or an odd function.

Common Functions

Definition

A linear function is a function of the form $f(x) = mx + b$, where m and b are constants. The constant m is the slope of the linear function and the larger m is in magnitude, the steeper is the graph of f. Its graph is a line with slope m and $y-$intercept b.

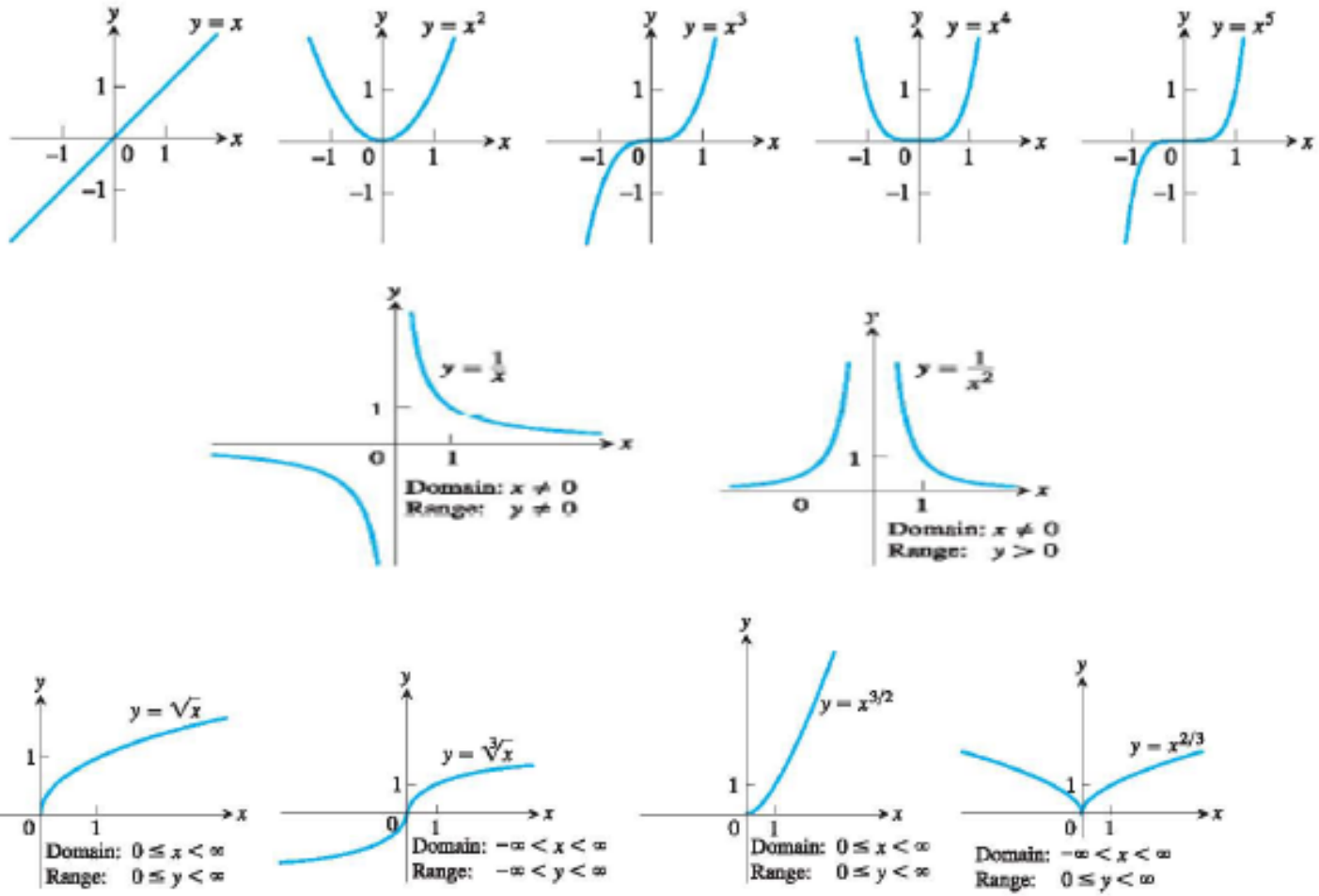
- **Identity function:** when $m = 1, b = 0, f(x) = x$
- **Constant function:** when $m = 0, f(x) = b$



Common Functions

Definition

A power function is a function of the form $f(x) = x^a$, where a is a constant.



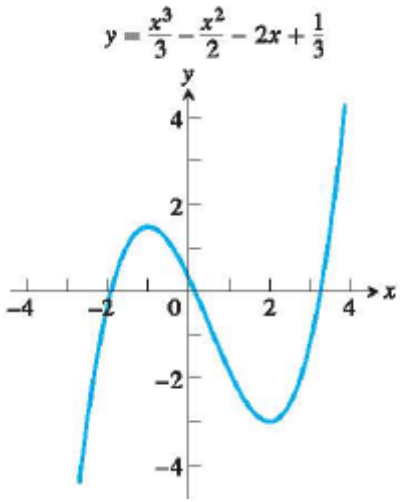
Common Functions

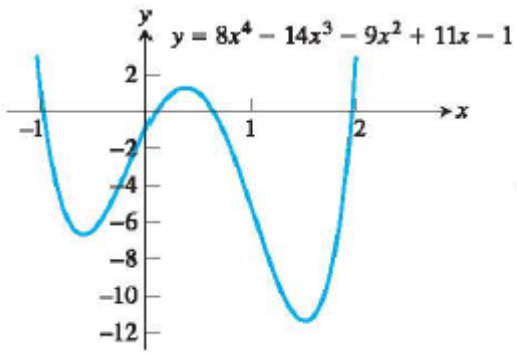
Definition

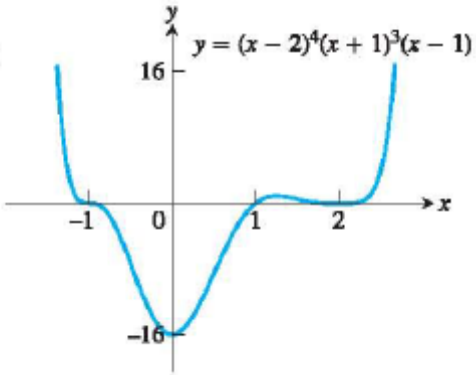
A polynomial function: A function p is a polynomial if $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ where n is a nonnegative integer and the numbers $a_0, a_1, \ldots, a_n$ are constants called the **coefficients** of the polynomial.

- Domain $= \mathbb{R} = (-\infty, \infty)$ for all polynomials
- If $a_n \neq 0$, n is called the **degree** of the polynomial

$n = 1$, it is called a linear function
$n = 2$, it is called a quadratic function
$n = 3$, it is called a cubic function



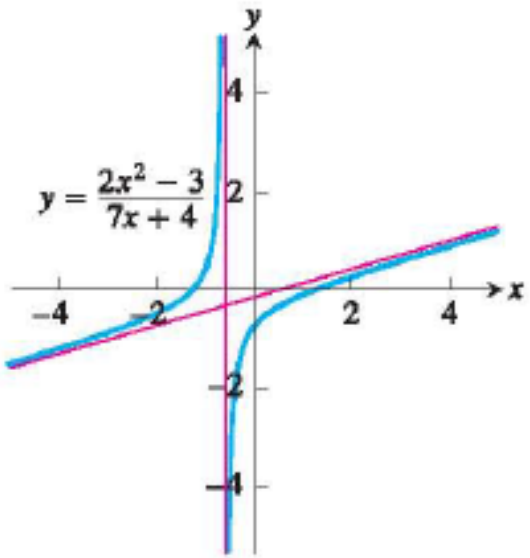


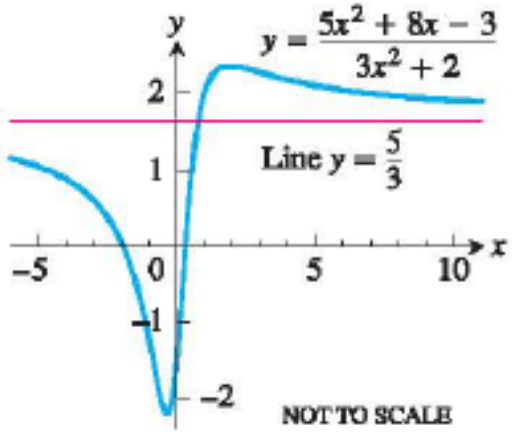


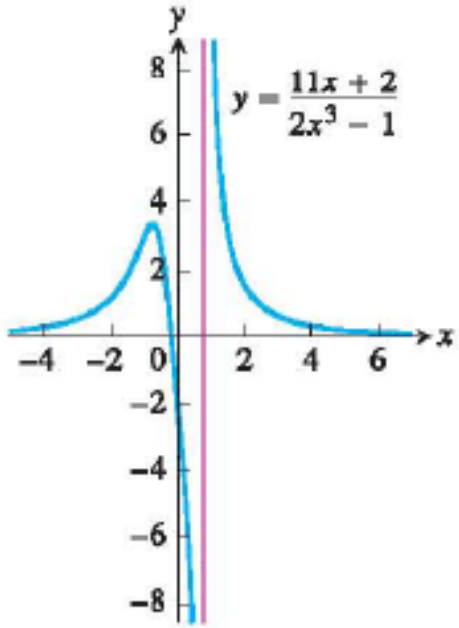
Common Functions

Definition

A rational function is a quotient (or ratio) of polynomials $f(x) = \frac{p(x)}{q(x)}$ where p and q are polynomials. The domain of a rational function is the set of all real x for which $q(x) \neq 0$.



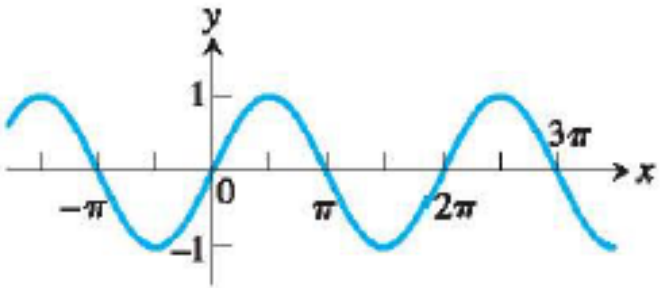




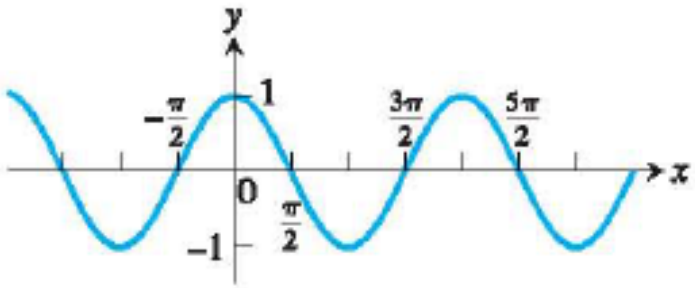
Common Functions

Definition

Trigonometric Functions: The six basic trigonometric functions are reviewed in Section 1.3. The graphs ofthe sine and cosine functions are shown below.



(a) $f(x) = \sin x$

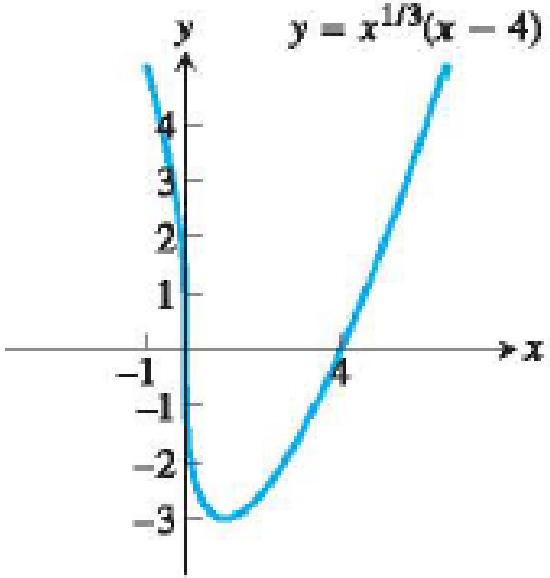


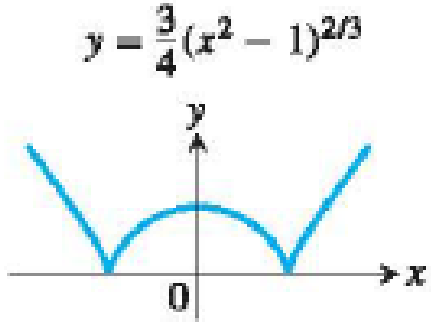
(b) $f(x) = \cos x$

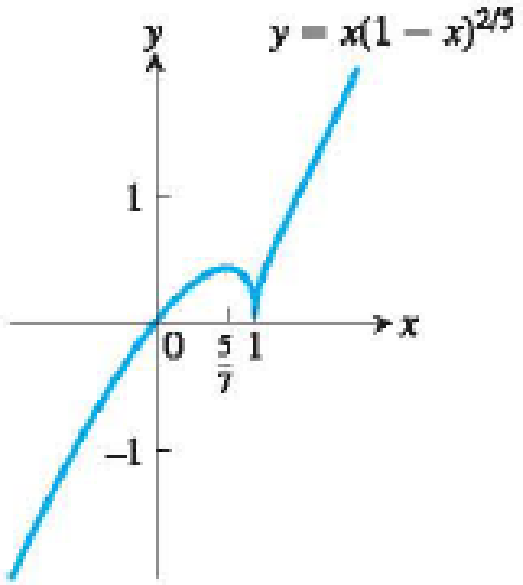
Common Functions

Note: transcendental functions

We will learn how to graph polynomials and rational functions in Chapter 4. These functions lie in the broader class of functions called algebraic functions. Any function constructed from polynomials using the algebraic operations of addition, subtraction, multiplication, division, and taking roots is an algebraic function. This class is in contrast to nonalgebraic functions, or transcendental functions, such as logarithms, exponentials, and trigonometric functions.



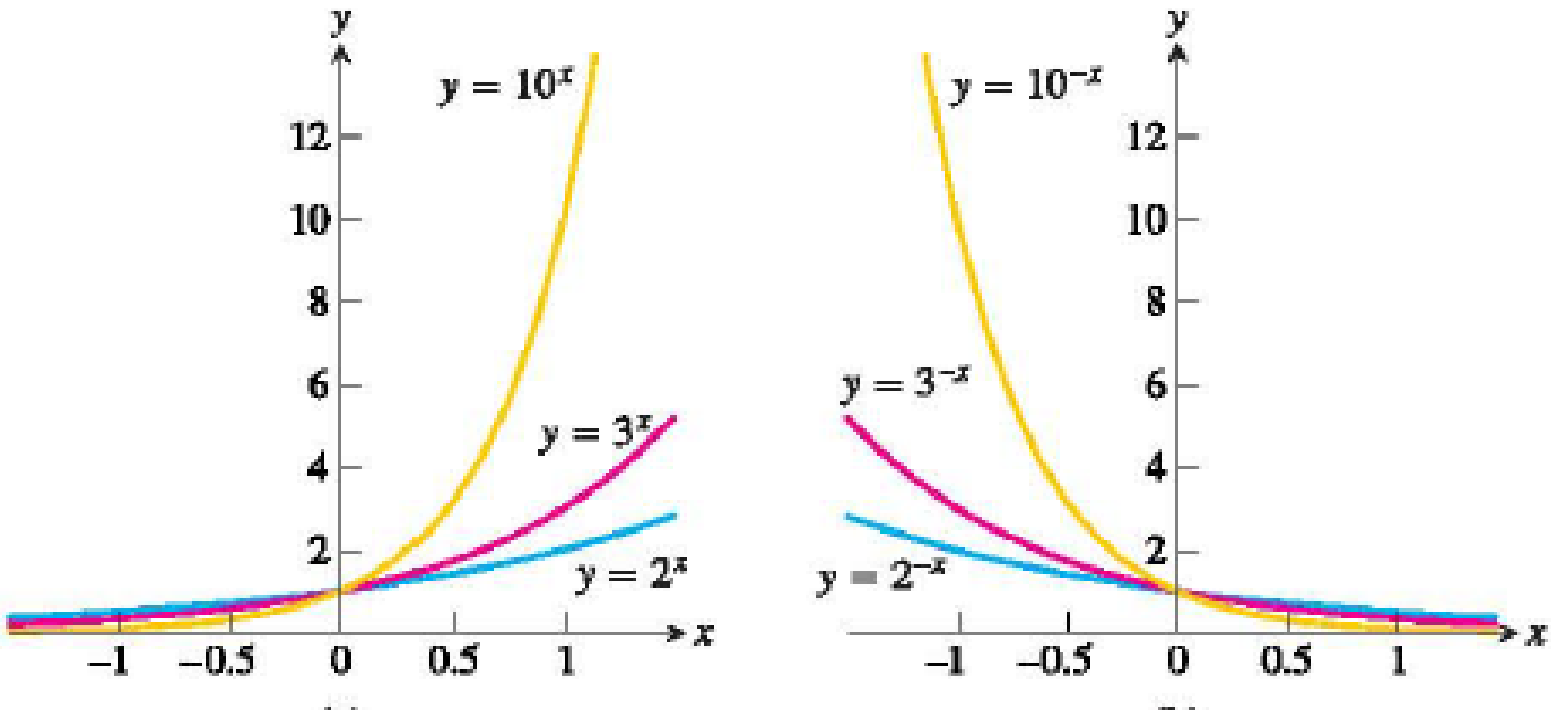




Common Functions

Definition

An Exponential function: A function of the form $f(x) = a^x$, where $a > 0$ and $a \neq 1$, is an exponential function. (Exponential functions will be more thoroughly explored in Chapter 7.)



End of Section 1.1.

Thank you.!!!